

Advanced Statistics in

Bernd Figner
bernd.figner@ru.nl

Week 1: Sept 8, 2025

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Before we start...

- Can everybody hear me and see the slides?
- You can always interrupt me: If I am too fast (or too slow), if something is unclear, if you have a question, etc
→ **interrupt me!**
- Who has/has not yet used R or RStudio?

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Overview

- **Some organizational things**
- **What is R? Why do we learn it? What is RStudio?**
- **Take your first steps in R/RStudio**
 - The different windows in R: Console, Editor, Graph (quartz)
 - Create and save an R/RStudio script
 - Install and load packages (aka libraries)
 - Variables, matrices, and data frames
 - Import, generate, and save data
 - Some first important functions and commands
- **Homework** (you can/should start in the lab session)

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Organizational Things

2 teachers, 2 coordinators, 1 teaching assistant

- **Bill Burk**; bill.burk@ru.nl (or william.burk@ru.nl, both work): teacher/coordinator
- **Marta Gomez Vargas**; marta.gomezvargas@ru.nl: teaching assistant
- **Bernd Figner**; bernd.figner@ru.nl: teacher/coordinator

In case you need to contact us via email, best to **always send/cc to all 3** of us.

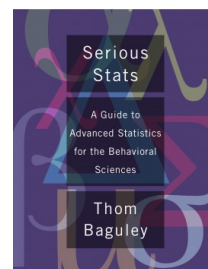
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Organizational Things cont.

Baguley, T. (2012). *Serious stats: A guide to advanced statistics for the behavioral sciences*. Basingstoke: Palgrave Macmillan.

- see <https://seriousstats.wordpress.com/2012/02/01/intro/>
- lectures follow roughly order of book chapters
- ideally, read the relevant chapter(s) **before** class
- things still unclear **after** class?
→ book good source for additional explanations
- it is not always the easiest and most entertaining book to read...



Additional reading materials will be announced later

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How this course works

- **Lecture** (10:30-12:15; recorded)
 - until Oct 27 MM 0.035; **from 10 Nov on EOS N 01.150**
 - Ask questions, give feedback!
- **Lab sessions** (13:30-15:15; 15:30-17:15); MM 2.110
 - Sign up for time slot
Brightspace → Administration → Groups; [Problems enrolling?](#)
 - Bring your own computer
 - Work on HW (Marta/Bill/Bernd there for help!);
questions/comments about lectures, R, ...
 - **Deadline** to upload HWs to Brightspace (Pitch2Peer): **10:20 AM**
on day of next class, typically (for exceptions, see syllabus)
 - **HW peer-grading:** in next week's lab session (everybody needs
to peer-grade a HW of another student!)
[peer-grading deadline: 1 week after HW deadline](#) (exceptions in
syllabus)

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Brightspace

- **now:** Syllabus, Discussion Boards, Week 1, ...
- **Things will be continuously updated and added**
 - Lecture slides
 - Homework instructions and example solutions
 - Example R scripts
 - List with literature and other resources
- **Discussion Boards: for now 2 boards**
 - Administrative things (course organization etc)
 - Content-related things
 - Post your R/Stats questions there
 - Answer each others' questions! ("incentive-compatible")

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Grades

- **Final exam** (planned for January 19, 2026): **80%**
 - Multiple choice
- **12 homework assignments: 20%**
upload them BEFORE the next class (deadline 10:20 AM)
 - R scripts, figures, write-ups; peer-graded as pass/fail
 - Your HW is counted as pass, only if you also peer-grade someone else's homework → deadline: 1 week after HW upload deadline
 - Final HW grade proportional to your total number of passed HW assignments
- **Brightspace participation**
 - Asking and answering questions
 - "Bonus" for active participation

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Grades cont.

- All grading components must be a minimum of 5.5
- If you fail the final exam
 - Teachers will give you opportunity for one retake
 - Maximum grade for retake: 10
- If you fail the homework grading component
 i.e., fewer than 7 out of 12 homework assignments passed
 - Teachers will give you opportunity for an extra assignment
 - Maximum grade for the homework grading component: 6

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Course Goals and Objectives

Course goals: Prepare students...

- for subsequent statistics courses
- their (minor and major) theses
- for a life as an independent academic researcher

After successfully completing this course...

...you should have a good **theoretical** understanding of statistics
AND at the **practical** level be able to use R for all (or nearly all)
your statistical needs

(more detailed goals and objectives in syllabus)

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Prerequisites

- **Not** required to be familiar with R and/or RStudio
- Familiar with basic statistics (such as the content of bachelor statistics courses) -> **poll**

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Course Content

- Hands-on „how to“ in R
- Theoretical background

→ Week-by-week schedule in syllabus

- Tentative, things might change a bit
- Let's have a look...

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Course Schedule

Class	Date	Topic	Book Chapter	HW due	Feedback due
1	08-09	Introduction to and first steps in R	-		
2	15-09	Samples, statistics, distributions	1,2	HW 1	
3	22-09	Graphs and figures	-	HW 2	HW 1
4	29-09	Confidence intervals & significance tests	3, 4	HW 3	HW 2
5	06-10	Linear regression	5	HW 4	HW 3
6	13-10	Correlation & covariance	6	HW 5	HW 4
7	20-10	Effect size & statistical power	7, 8	HW 6	HW 5
8	27-10	Exploring and dealing with messy data	9, 10	HW 7	HW 6
	03-11				HW 7
9	10-11	Bayesian inference	11		
10	17-11	Multiple linear regression	12	HW 8	
11	24-11	ANOVA & ANCOVA	13	HW 9	HW 8
12	01-12	Logistic regression	14	HW 10	HW 9
13	08-12	Interactions	15	HW 11	HW 10
14	15-12	Contrasts	16		HW 11
15	05-01	Repeated-measures (M)AN(C)OVA	17		
16	12-01	Exploratory factor analysis: wrap-up	-	HW 12	
	19-01	Exam (Comenius building 12:45-14:45)			

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- **Free, open-source** programming language/environment
 - Statistics
 - Graphics
- Originally developed by 2 stats profs in New Zealand
<http://www.nytimes.com/2009/01/07/technology/business-computing/07program.html>
- Very flexible and highly extensible
- Under active development
 - Mainly by statisticians working in academia
- Widely used in academia and business
 (e.g., Google, New York Times, ...)

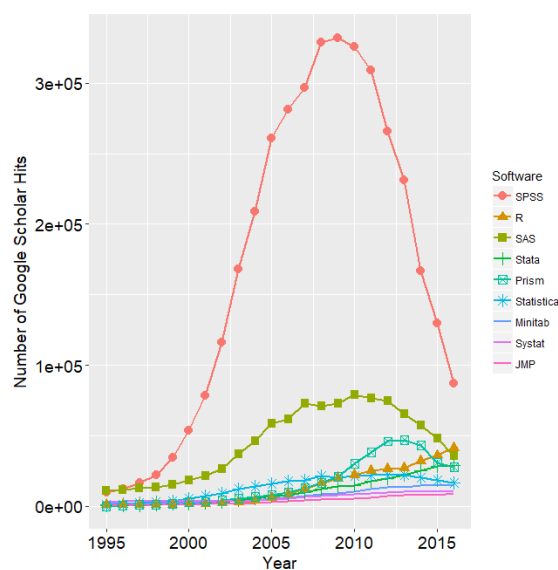
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R is the future...

<http://r4stats.com/articles/popularity/>

R took over SAS in 2015 and might overtake SPSS ...



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- **Available for PC, Mac, Linux**
- **Base installation:** basic stats, plots, and other functions
- Additional modules for different purposes
→ "**packages**" (aka "libraries")
- **Packages for**
 - More basic statistics (correlations, regressions, ANOVA, ...)
 - Advanced methods (e.g., mixed models; SEM; Bayesian; ...)
 - Graphics and figures
 - Newest state-of-the-art methods: quickly available
 - Can be a bit confusing in the beginning (> 25'000 packages)
- **Not only statistics**
 - Very versatile and powerful environment, including data management, computations, simulations, publication quality plotting, ...

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BUT

- Not point'n'click... (although: Deducer; R Commander; ...)
- It is a **programming environment** (similar to Python, Matlab, etc)
- You will write **scripts** (i.e., code/commands that tell R what computations it should do)
- It takes some getting used to...
- **Think like a programmer**
 - Each line of code tells the computer (R) what to do
 - Making errors is common
 - Finding and fixing errors is an important part of working with R
 - **Run and check each line of code, line after line**
 - **No really: Run and check the result of EACH line of code, immediately after you wrote it!**

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R versus RStudio

- **R** consists of „free-floating“ windows, including
 - Console
 - Script window (editor)
 - Graphs
 - Help
 - **RStudio**: a nice GUI (graphical user interface)
 - Windows not free-floating, but neatly arranged into 1 window
 - Data viewer (and a few more nice things)
 - R and RStudio are free (there is a commercial RStudio version, too)
 - Ultimately, RStudio is mostly a pretty surface for R
 - (Virtually) everything you do in RStudio, you can do in R, and vice versa
- **We (and the book) talk about R, but you can do everything the same way in RStudio**

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Installation of R and RStudio

- First install R
- If you want RStudio, install RStudio next
- For both: install the **newest** version
- Installation is very easy (e.g., search for "R" or "RStudio")
 - Instructions for R installation: see homework videos
 - Instructions for RStudio installation: see homework videos
- To start the program
 - If you use R: start R
 - If you use RStudio: start RStudio (no need to start R)

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HELP!

If you get stuck: **Error in ...**

- Typo in your syntax? (most common reason for problems)
→ **check each line of code right after you wrote it!**
- Forgot the syntax for a specific command?
e.g., for command `mean()`
→ **run the command ?mean**
- Starts R's built-in help documentation
- alternative commands that do exactly the same
`?mean()`; `help(mean)`; `help(mean())`
- **Brightspace** → `CommandsFunctions_FirstFewWeeks.docx` and `short-refcard.pdf`

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Still HELP!

- Very very active R community!
- Use your preferred search engine, e.g.:
 - "R compute mean of variable with missing values?"
 - Copy/paste **error message** into search engine
- **Particularly good answer sites**
 - stackexchange (aka cross validated), stackoverflow
 - Quick-R
 - R Email list archives
- LLM-based tools like ChatGPT
- Post problem on Brightspace (Discussion Board)
 - Be **as precise as possible** (copy/paste **your code and error**)
 - Try to answer each others' questions!

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In R...

... many roads lead to Rome

- Typically, several ways to achieve the same thing
- They can be all correct and equally good
- Don't be confused if you see different solutions for the same problem

... **error** ≠ **warning**

- **error** → I couldn't do what you asked me to do
- **warning** → I did something but I'm not sure it's OK (might be all fine, but you should check)

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Naming things

- Don't use spaces in names of variables, data frames, files, or folders
 - **Don't:** "My R Script.R"
 - **Good to use:** "My_R_Script.R"
 - **Good to use:** "MyRScript.R"
 - same for folders on your hard drive etc

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Let's switch to R

The R script will be on Brightspace!

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Welcome back to PowerPoint

The R script I just used will be on Brightspace!

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What we just saw

- Console, Editor, Graphics windows
- No data window (unlike SPSS, excel, etc)
RStudio does have kind of a data window, R does not (well, kinda)
- Instead use:
 - commands: `head()`, `tail()`, `mydata[1:30,]`
 - Excel, OpenOffice, ... → before opening, make a separate copy of the file; Excel changes the file even when you don't save it!
- Basic important R commands
- List of basic commands from the first week on Brightspace
(class, example scripts, book):
→ `CommandsFunctions_FirstFewWeeks.docx`

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Set the Working Directory

- Create a folder; place the data files in that folder
- After starting **R**, change the working directory to that folder

Mac

```
setwd("~/Stats_New_I_II/2025_2026/Week_01")
```

PC (both / or \\ work)

```
setwd("C:/Stats_II_IntroR/2025_2026/Week01/")
```

```
setwd("C:\\Stats_II_IntroR\\2025_2026\\Week01\\")
```

Check whether setting the directory worked: `getwd()`

Do not use spaces in names of folders and scripts!

Copy/pasting from my slides does not always work:

- correct quotation marks: " and '
- doesn't work: “ ” ’

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How to get this file path?

- **R on Mac:** drag and drop folder into the R script
- **RStudio on Mac:** drag and drop into "Terminal" app; copy path from there
- **PC:** In File Explorer window, go to the file/folder you want the path for, then you can do one of 3 things:
 - click once in the window that shows the path; copy that to R/RStudio
 - right-click on the object and then open the "Properties" window; copy that to R/RStudio
 - shift right-click and choose copy path

→ Do not forget to replace each \ with either / or \\

→ Check with `getwd()` to see whether it worked

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Importing Data

Recommended

- Non-proprietary formats, e.g., **csv** (Comma Separated Values)

```
MyData1 <- read.csv("ThesisData.csv")
```

```
MyData1 <-  
read.csv("C:/Documents/data/ThesisData.csv")
```

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Importing Data

NOTE: Dutch OS uses decimal comma, instead of decimal point!

- After importing, check data using `head(mydata)` and `tail(mydata)`
- If it doesn't look right, try some of the following:
`read.csv("ThesisData.csv", sep = ";")`
`read.csv2("ThesisData.csv")`
`read.csv2("ThesisData.csv", sep = ";")`
`read.csv2("ThesisData.csv", sep = ",")`
`...`

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SPSS and Excel Files

- Export from SPSS/Excel as csv file ("save as..." or "export..."), then in R use `read.csv()`
- Use specific R libraries to read .xls or .sav files; e.g., `library(foreign)`

```
MyData1 <- read.xls("ThesisData.xls")
```

```
MyData1 <- read.spss("ThesisData.sav",  
  to.data.frame = TRUE)
```

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Important

Always check thoroughly whether import worked (from .csv, from .xls, from .sav, ... ALWAYS!)

- Show whole data frame in Console: simply run command like `MyData1`
- `nrow()`, `ncol()`, `head()`, `tail()`, ...
- Missing values in R: **NA** (not "99999" or "." etc)
- Important to check not only the first few rows of data!
Last row(s) sometimes contain all NA entries, especially when you import a proprietary format such as .sav or .xlsx
→ such empty rows need to be removed

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Saving Data Files

Recommended: Non-proprietary formats, e.g., **csv**

```
write.csv(MyData2, file = "ThesisData2.csv",
row.names = FALSE)
```

```
write.csv(MyData2, file =
"C:/Documents/data/ThesisData2.csv", row.names =
FALSE)
```

row.names = FALSE

Otherwise, R adds a very first column with the numbers 1, 2, 3, etc

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Saving Data Files cont.

Not recommended (but possible): Proprietary file format

For example Excel

```
library(dataframes2xls)
write.xls(MyData2, file = "ThesisData2.xls")

library(xlsx)
write.xlsx(MyData2, file = "ThesisData2.xlsx",
sheetName = "Demographics")
```

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Data Frames

The main object type for your data

- **Columns:** Variables (pp_code, f_gender, age, ...)
 - **My recommendation:**
 - Use variable names like **f_...** (for factors), **o_...** (for ordered factors), **n_...** (for numerical), etc
 - Your future self will thank you
- **Rows:** Data units; e.g.
 - Participants (e.g., SPSS) → **"wide format"**
 - Repeated measures (trials) of a within-subject design
 - **"long format"** **[next week!]**

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Data Frames

Indexing: `MyThesisData1[rows, columns]`

- `MyThesisData1[1, 1]`
- `MyThesisData1[, 3]`
- `MyThesisData1[2:5, c(1,3)]`
- `MyThesisData1[2:5, c("pp_code", "f_gender")]`

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Homework/Assignments

All the following information (with more detail) is also in the file **HW_Instructions_Week01** on Brightspace -> Week 1 -> Homework

(a) Install R (and RStudio if you want) on your computer

Instructions, if you need them (rather old videos, but still helpful)

R

<http://www.youtube.com/watch?v=W101Mof0Exc&index=7&list=PLycnP7USbo1V3jlyjAzWUB201cLxPq4NP>

R and RStudio

<http://www.youtube.com/watch?v=WJDrYUqNrHg>

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(b) Go through my example script (on Brightspace): [AdvStatsR_Week_1_ClassScript.R](#)

- Do all the things that I do; do not just copy/paste, but try out your own variations (e.g., create your own, longer vector etc)

(c) Watch two videos by Dan Goldstein

He shows useful basics (some of it, we'll be covering only later)

- <https://www.youtube.com/watch?v=ZmtkqaRVTDC>
- <https://www.youtube.com/watch?v=GcmorHXygoI>

Try out the commands/analyses (plots, etc) that Dan does

Important things to remember from these videos

- how to read an external data file (including setting working directories on a Windows PC)
- use of \$ (as in: dataframe\$variable)
- creating relatively simple plots (advanced ones: later classes)
- creating a simple linear regression (but we'll cover that later!)
- how to get help using ?

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(d) Do some computations, descriptive statistics, load and save data, etc

- Things like loading file [parenting.sav](#); compute the mean of the variable Steps, etc...
- Do in R all the things that the document asks you to do

Upload your homework to Brightspace (for peer-grading next week)

- Copy/paste your relevant R code into the word document with the homework instructions
- Save your completed homework; turn this document into a pdf file with **your name** in the file name (HW1_BerndFigner.pdf)
- Upload this file to Brightspace -> Week 1 -> Pitch2Peer (as "slideshow pitch")
- **Deadline: Monday Sept 15 10:20!**

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(e) Not mandatory, but highly recommended (Baguley book)

For next week's class: Chapters 1 and 2

- **Chapter 1**
 - Samples and populations; sample versus population parameters
 - Descriptive versus inference statistics and statistics
 - Different types of data: discrete versus continuous
 - Measures to describe the central tendency in data (e.g., mean, median)
 - Measures to describe the dispersion in the data (e.g., variance, SD)
- **Chapter 2**
 - Common probability distributions for discrete and continuous data
 - Skew, kurtosis
 - Central limit theorem

At the end of each chapter, there is a section with R code; I recommend that you do all these things yourself in R!

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If you encounter problems, have questions, etc:

- Ask in the lab session!
- Put them on the Discussion Board in Brightspace
- Feel free to answer others' questions!

Questions or comments?

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Important things covered this week

- Basic R functionalities: Console, Editor, Graph, Help window; R scripts; installing/loading R packages
- Variables, matrices, data frames
- Loading, generating, and saving data frames/data files
- Indexing in data frames
- Important first/basic R commands (setting working directory, etc)

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Thank you for the attention!

**See you in the lab session in the
afternoon**

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